Jonah Rosenblum

Ann Arbor, MI — jonaher@umich.edu — jonahrosenblum.com

Research Interests

My goal is to protect sensitive data, even when users and software cannot be fully trusted. I build secure systems that enforce data-use policies using context (time, location, and code properties), not only identity or credentials. My work also spans genomic privacy, Rowhammer defenses, and scalable memory freshness.

Education

Ph.D. in Computer Science and Engineering (CSE) *2022–present*
University of Michigan, Ann Arbor, MI; advisor: Satish Narayanasamy; GPA: 3.99

M.S., CSE (2021) — **B.S. in Computer Science (2017–2020)**
University of Michigan, Ann Arbor, MI — GPA: 3.83

Experience

Hewlett Packard Enterprise Labs, CA *May 2025–present*
Research Associate, Systems Architecture Lab
Initiated formal verification to detect security-intent misconfiguration; redesigned DAOS security for a new authentication backend.

Google, Cloud Infrastructure, CA *May–Aug 2021*
Software Engineering Intern
Analyzed network/application priority inversions across clusters; recommended QoS for less latency-sensitive traffic.

Google, Cloud Trace, Waterloo *May–Aug 2020*
Software Engineering Intern
Implemented graceful shutdown in OpenTelemetry to ensure all traces and metrics are exported.

Publications

- **Jonah Rosenblum**, Juechu Dong, Peter Chen, Satish Narayanasamy. "Timelock Drive: Isolated Time-Based Defense for Storage Systems." *OSDI*, 2026.
- **Jonah Rosenblum**, Juechu Dong, Satish Narayanasamy. "Confidential computing for population-scale genome-wide association studies with SECRET-GWAS." *Nature Computational Science*, 2025.
- Juechu Dong, **Jonah Rosenblum**, Satish Narayanasamy. "Toleo: Scaling Freshness to Terascale Memory using CXL and PIM." *ASPLOS*, 2024.
- Kevin Loughlin, **Jonah Rosenblum**, Stefan Saroiu, Alec Wolman, Dimitrios Skarlatos, and Baris Kasikci. "Siloz: Leveraging DRAM Isolation Domains to Prevent Inter-VM Rowhammer." *SOSP*, 2023.

Technical Skills

Programming: C, C++, Python, Go, Rust — **Systems:** Linux/QEMU
Formal verification: Dafny, Hoare logic

Recognition & Service

- 2023 Outstanding GSI Award; one of three recipients among 400+ student instructors.
- 2025 USENIX ATC Artifact Evaluation Committee.
- Invited presentation at the Young Architect Workshop (YArch).